

Predicting Supply Chain Disruptions and Recovery Times Using Machine Learning

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Introduction

Background and Context

Many companies nowadays need very intricate supply chains because they include suppliers, transportation systems, facilities for production, and markets that vary by region. One thing about these supply chains is that when one part of it is disrupted, the effects of that spread very quickly and cause all sorts of delays. These can lead to higher costs, loss of revenue, and decrease customer satisfaction. Some examples of disruption include cyber-attacks, labor strikes, incidents at factories, geopolitical issues, congestion at ports, and many other types of problems that can arise in the operational process.

This project focuses on the big business problem of predicting supply chain disruption recovery time. Rather than relying only on the reactive type of strategies after a disruption has already happened, businesses can use predictive analytics to prepare for possible future disruptions, estimate the duration of the recovery of them, and see the factors that make disruptions more severe in the first place. This type of analysis shows the importance of business planning, allocation of resources, management of suppliers, and the reduction of risk altogether.

Objectives of the Project

The main goal of this project is to study supply chain disruption data and build a machine learning model that can help predict the full recovery time. The analysis done from this also should identify the variables that affect recovery duration as the strongest. The project highlights multiple research questions, such as:

- Which variables when it comes to operations and logistics drive supply chain disruption outcomes?
- How does disruption type relate to recovery time?
- How useful is machine learning for estimating the risk of recovery?
- Which variables are the most important for the model predictions?

The final goal of all of this is to understand what the connections are between the learning results and how they apply in a real-world business context. By learning which factors drive long recovery periods the most, supply chain managers can plan better regarding backup suppliers, strategies for disruption responses, risk planning, and investments on preventative measures so disruptions don't affect the business too severely.

Methodology

Dataset Description

The data for this research study, the Supply Chain Disruption and Recovery dataset, is sourced from Kaggle. The original dataset contains 100,000 supply chain disruption events across 5 industries: automotive, electronics, pharmaceuticals, consumer goods, and aerospace with 15 variables of analysis. The disruptions are tracked across multi-tier supplier networks ranging from Tier 1 to Tier 4. These disruptions are categorized into 6 types: natural disasters, cyberattacks, geopolitical issues, labor strikes, port congestion, and factory incidents. This structured dataset includes a variety of operational, logistical, and event-based variables that track interruption events and their subsequent recovery results. The dataset contains both numerical and categorical data because of the variety of variables, which makes it very ideal for machine learning applications.

It is designed to support both classification (predicting if a disruption occurs) and regression (estimating recovery time) tasks. Overall, this dataset has a strong foundation for analyzing the main causes of supply chain disruptions and what recovery timelines can look like. This project ultimately aims to highlight meaningful patterns and relationships that can support data-driven decision-making in supply chain risk management as well as other business settings as well. The following table categorizes these variables into their respective natures: Operational, Logistical, Event-based, and Target.

Table 1: Data Variables and Nature

#	Variable Name	Description	Feature Nature
*	full_recovery_days	Number of days to achieve full operational recovery (The target variable).	Target
0	disruption_id	Unique identifier for the disruption record.	Non-Predictive
1	partial_recovery_days	Number of days to achieve partial operational recovery (Excluded to prevent leakage).	Non-Predictive
2	disruption_type	Type of disruption event (e.g., Cyber Attack, Factory Incident, Labor Strike).	Event-based
3	industry	The industry sector of the affected supply chain (e.g., Electronics, Pharmaceuticals).	Operational
4	supplier_tier	Tier level of the supplier affected (1, 2, or 3).	Logistical
5	supplier_region	Geographic region where the supplier is located.	Logistical
6	supplier_size	Size category for supplier (Small, Medium, Large).	Logistical
7	has_backup_supplier	Indicator if a backup supplier is available (True/False or 0/1).	Logistical
8	disruption_severity	Severity score rating of the disruption event (Scale 1–5).	Event-based

9	production_impact_pct	Percentage reduction in production capability due to disruption.	Operational
10	revenue_loss_usd	Estimated financial revenue loss in USD resulting from the disruption.	Financial
11	response_type	Mitigation strategy implemented (e.g., Alternative Supplier, Customer Delay).	Operational
12	response_time_days	Time taken in days to execute the mitigation response strategy.	Operational
13	permanent_supplier_change	Indicator if the disruption led to a permanent change in vendor (True/False or 0/1).	Logistical
14	severity_impact	Interaction features combining disruption severity and production impact percentage.	Engineered

Key Variables for Analysis

For the purpose of this study, the key variables utilized to build the predictive machine learning framework include:

- disruption_severity (intensity of event)
- has_backup_supplier (redundancy indicator)
- revenue_loss_usd (financial impact)
- severity_impact (derived impact score)

- production_impact_pct (operational loss)
- response_time_days (reaction speed)
- supplier_tier (supply chain depth)
- response_type (mitigation strategy)
- industry (sector context)
- full_recovery_days (recovery duration)

Analytical Techniques

1. Data Preprocessing The preprocessing phase focused on transforming the raw Kaggle dataset into a machine-learning-ready format:

- **Data Cleaning:** Verification of null values and data types to ensure consistency across the 100,000 records.
- **Feature Engineering:** Creation of the severity_impact variable, an interaction term derived by multiplying disruption_severity and production_impact_pct to capture the compound effect of an event.
- **Label Encoding:** Used for ordinal or multi-class features such as industry, supplier_region, and supplier_size.
- **Feature Scaling:** Implementation of StandardScaler on continuous numerical variables (e.g., revenue_loss_usd, production_impact_pct) to normalize the data range.
- **Data Splitting:** A 80/20 train-test split was used to evaluate model generalizability.

2. Exploratory Data Analysis (EDA) EDA was conducted to identify underlying patterns and validate the project's assumptions:

- **Correlation Analysis:** A heatmap was generated to identify the strongest relationships between features and the target `full_recovery_days`.
- **Univariate & Bivariate Analysis:** Visualizing distributions of recovery times and comparing them across different industries and supplier tiers.
- **Outlier Detection:** Identifying extreme disruption events; it was determined that these outliers represented valid high-impact scenarios and were retained to ensure the model could account for rare but severe crises.

3. Modeling and Evaluation

- **Primary Model: Random Forest Regressor.** This ensemble learning method was selected for its ability to handle non-linear relationships and provide feature importance insights.
- **Evaluation Metrics:** The model's performance was measured using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R-squared (R^2)** to assess how well it explained the variability in recovery times.
- **Feature Importance:** A statistical analysis of which variables (e.g., `disruption_severity`) had the most significant impact on the model's predictive accuracy.

Tools and Technologies

Platforms

- **Python:** The primary programming language used for all data manipulation and modeling.

- **Jupyter Notebook:** The interactive development environment used to document, visualize, and execute multi-stage analysis.

Key Libraries

- **Pandas:** For data structures and handling the .csv datasets.
- **NumPy:** For numerical operations and array of manipulations.
- **Scikit-learn:** Used for preprocessing (StandardScaler), model selection (train_test_split), implementing the Random Forest algorithm, and performance evaluation.
- **Matplotlib & Seaborn:** Used for generating visual analytics, including correlation heatmaps, feature importance bar charts, and distribution plots.

Executive Summary

Data Preprocessing

Data preprocessing is the process of cleaning and transforming raw data into a format that is suitable for building and training machine learning models. Real-world data is often “messy” which means that it can contain incomplete, inconsistent, or missing values. Effective preprocessing removes any incomplete or missing values and identifies outliers that may skew the results of the study. The removal of “noise” or irrelevant information allows the model to identify meaningful patterns. These are the steps for the Data Preprocessing used in this study:

1. **Importing required libraries:** We utilize libraries like Pandas for data manipulation, Scikit-learn for encoding and scaling, and NumPy for mathematical operations.

2. **Loading the dataset:** The raw data from the supply_chain_disruption_recovery.csv file is loaded into a data frame for analysis.
3. **Checking for missing or null values:** A scan of the 100,000 records confirmed that the dataset does not contain any missing or null values, ensuring a complete data foundation for the model.
4. **Dropping irrelevant columns:** Like the EDA phase, we drop the disruption_id column. Because this is a unique identifier for each event, it does not provide any predictive power for calculating recovery days.
5. **Feature Engineering:** This involves creating new variables to help the model understand complex relationships. In this case, we created severity_impact by multiplying the disruption severity by the production impact percentage. This helps the model to see how the intensity of an event compounds with the actual loss of capacity.
6. **Encoding Categorical Data:**
 - a. **Label Encoding:** We applied this to variables like supplier_size and supplier_region to convert text categories into numbers.
 - b. **One-Hot Encoding:** This was used for the response_type column, creating separate binary (0 or 1) columns for each strategy to ensure the model does not assume a mathematical ranking between different strategies like "Inventory Buffer" and "Customer Delay."
7. **Handling Outliers:** In this dataset, outliers represent extreme disruption events (e.g., very long recovery times). Unlike other studies where outliers are removed, we chose to retain them.

8. **Feature Scaling:** Continuous variables like `revenue_loss_usd` and `production_impact_pct` have very different scales. We used `StandardScaler` to normalize these features so that variables with larger numbers do not unfairly dominate the model's logic.
9. **Data Splitting:** We divided the data into a training set (80%) and a testing set (20%).

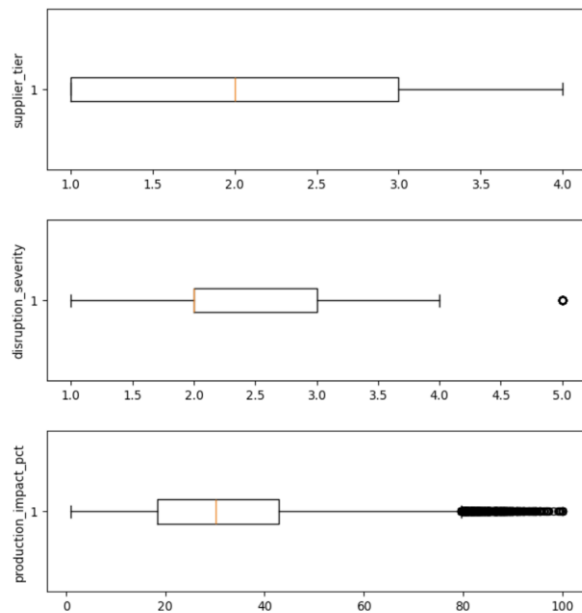


Figure 1: Supplier Tier, Disruption Severity, Production Impact Outliers Boxplots

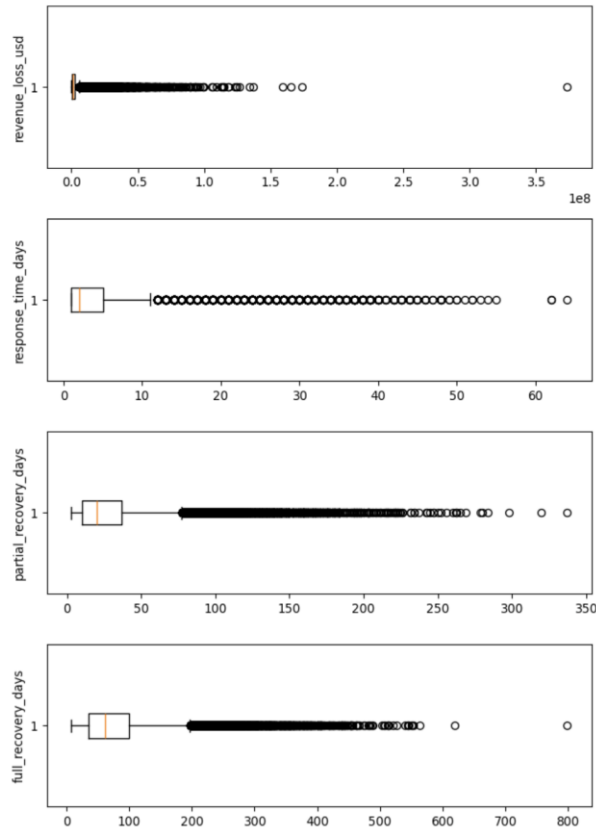


Figure 2: Revenue Loss, Response Time, Partial/Full Recovery Days Outliers Boxplots

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is understanding the data sets by summarizing their main characteristics, often plotting them visually. It's a foundational approach that uses statistical and graphical techniques to spot patterns and anomalies in a dataset and gain a deeper understanding (Komorowski et al., 2016). This step is very important, especially when we arrive at modeling the data to apply Machine learning. Plotting in EDA consists of Histograms, Box plot, Scatter plot, and many more. It often takes a lot of time to explore the data. Through the process of EDA, we can ask to define the problem statement or definition on our data set, which

is very important. These are the steps for the Exploratory Data Analysis (EDA) used in this study:

1. **Statistical Summaries:** We calculated descriptive statistics for all numerical variables, such as `disruption_severity`, `production_impact_pct`, and `full_recovery_days`. These summaries provide a quick overview of the data's central tendency (mean/median) and spread (standard deviation), helping us identify the "typical" disruption event versus extreme cases.

2. **Univariate Analysis:**

- a. For the numerical variables, there is a high skewed mean of about 2.5 mil but a maximum of 373 mil, which means that there is a lot of outliers and revenue loss.
- b. The impact of the production time is 31.4%.
- c. The recovery times take up to 27 days for partial recovery and 75 days for full recovery.
- d. The most frequent disruption time is "Port Congestion."
- e. The sector that is the most represented industry is "Electronics."
- f. The region that is the most dominant supplier region is "Asia-Specific."

3. **Bivariate and Multivariate Analysis:** By plotting `disruption_severity` against `full_recovery_days`, we could visually confirm that as the severity of a disruption increases, the number of days required for recovery also tends to rise.

- a. Numerical Trends: The boxplot reveals that Cyber Attacks and Geopolitical issues have a wider "interquartile range" for recovery times compared to Port Congestion. The port issues are more predictable while cyber incidents can range from quickly fixed to long-term failures.

- b. Categorical Trends: Pharmaceuticals and Aerospace industries show higher median revenue loss than Consumer Goods. This is most likely due to the higher value of goods and strict regulatory requirements.
 - i. Larger suppliers tend to have backup suppliers more frequently than smaller suppliers to add a buffer for total production failure.
 - c. Multivariate: Tier 1 suppliers consistently reside in the upper-right quadrant of the scatter plot. Tier 3 or Tier 4 suppliers might fail, but if a Tier 1 supplier fails, it could cause extremely high revenue loss.
- 4. **Correlation Analysis**: To understand how variables relate to one another, we generated a correlation heatmap. This allowed us to see which factors, such as `disruption_severity` or `revenue_loss_usd`, have the strongest statistical relationship with our target variable, `full_recovery_days`. A strong correlation suggests a potential predictor for our model.
- 5. **Histograms for Distribution**: We used histograms to plot the distribution of recovery times. These plots helped us visualize whether the data follows a normal distribution or is skewed, which informs our choice of modeling techniques.
- 6. **Box Plots for Outlier Detection**: Box plots were used to identify extreme disruption events. In our dataset, points appearing outside the "whiskers" of the box plots for variables like `revenue_loss_usd` represent high-impact, rare disruptions. Unlike standard data cleaning where these might be removed, we identified these as critical scenarios that must be understood for supply chain resilience.

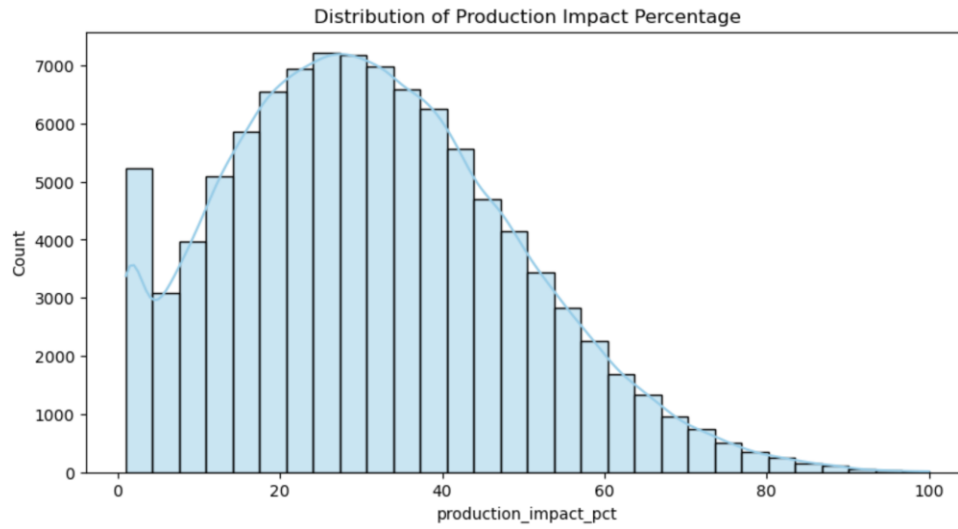


Figure 3: Distribution of Production Impact Percentage Histogram

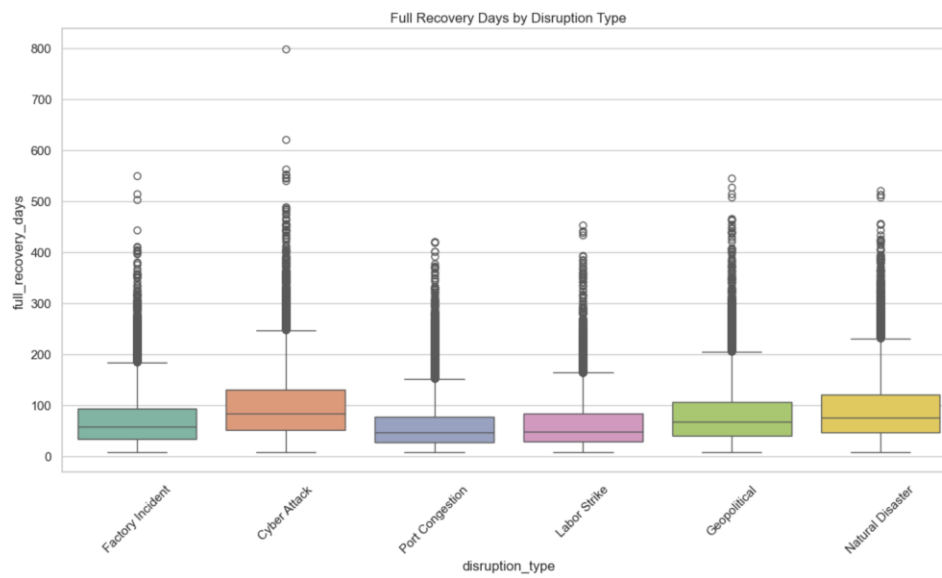


Figure 4: Full Recovery Days by Disruption Type Boxplots

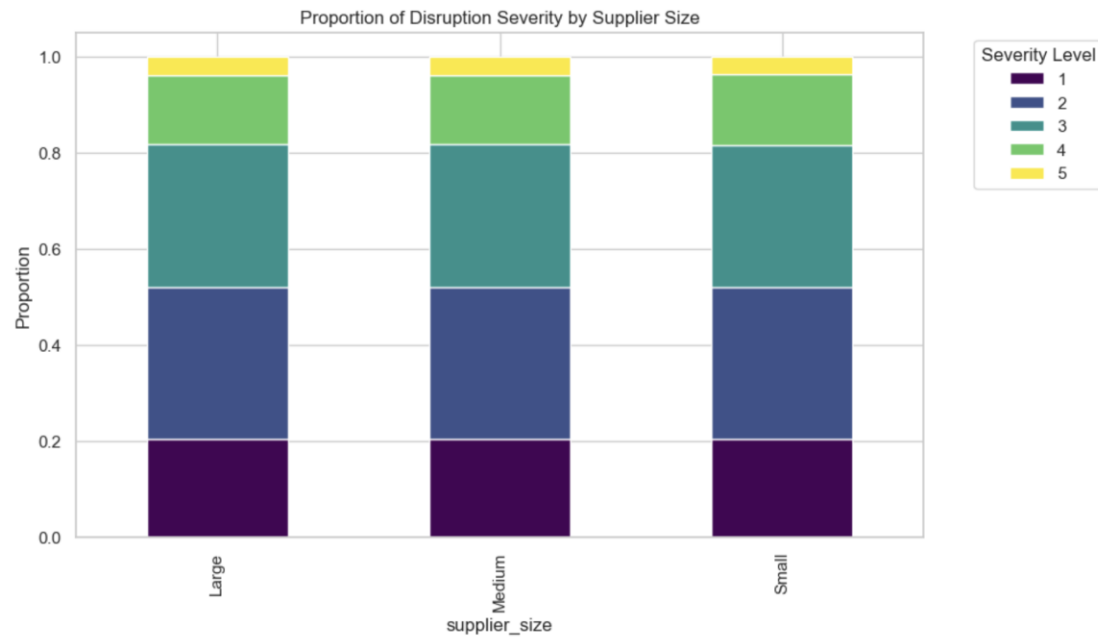


Figure 5: Proportion of Disruption Severity by Supplier Size Bar Chart

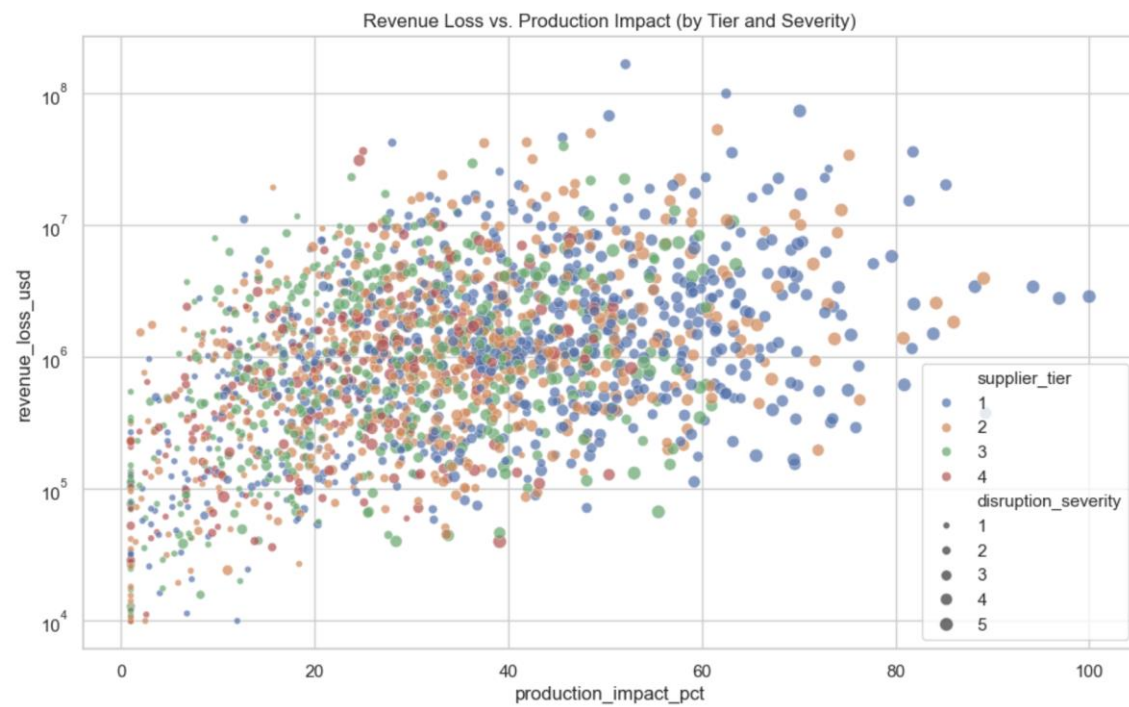


Figure 6: Revenue Loss vs Production Impact Scatterplot

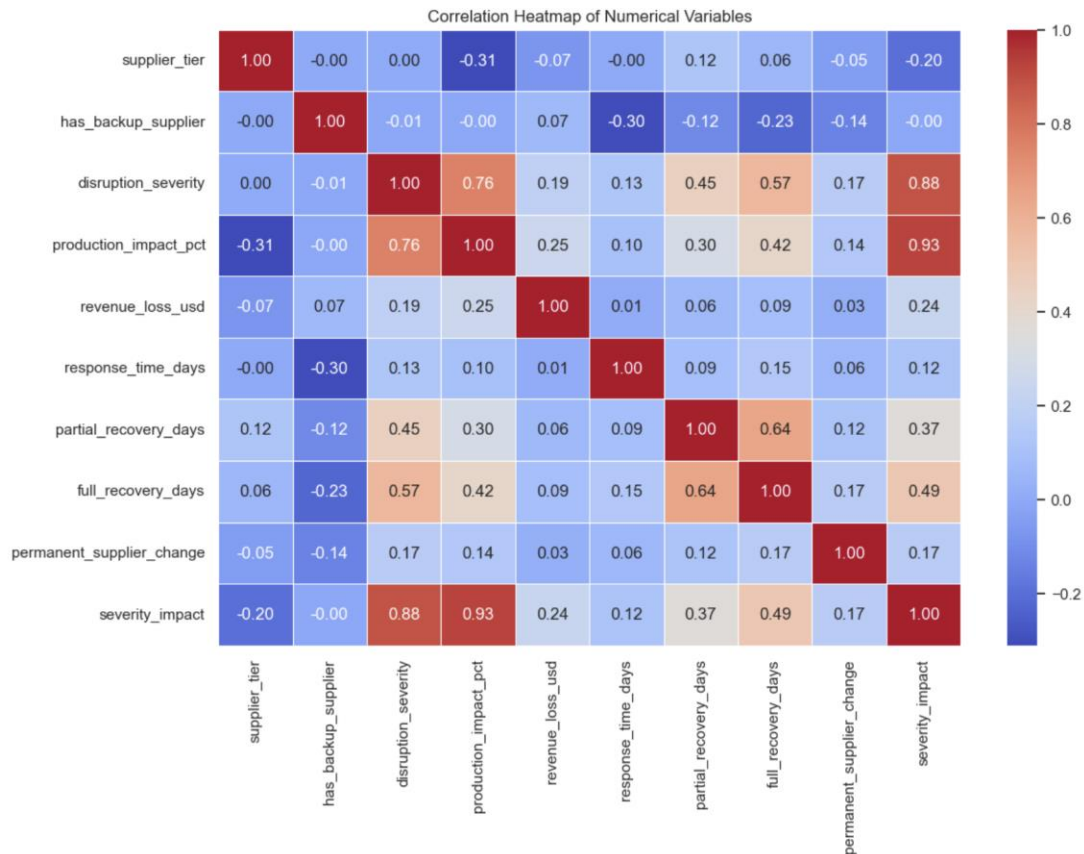


Figure 7: Correlation Heatmap of Numerical Variables

Key Insights from EDA

- Higher disruption of severity and production impact are strongly associated with longer recovery times.
- Revenue loss and recovery duration show significant variability, indicating the presence of high-impact disruption events.
- Certain disruption types consistently lead to longer recovery periods, which suggests industry-specific vulnerabilities.
- Response strategies vary across disruption types, indicating different approaches to managing supply chain disruptions.
- Outliers observed in the dataset appear to be valid in extreme cases rather than data errors and were ultimately retained for modeling.

Principal Component Analysis (PCA)

PCA was used to see whether the model predictors could be changed into a smaller set while keeping most of the information that is in the dataset. Before PCA was done, the categorical variables in the dataset were one-hot encoded, and the predictor matrix was standardized too. The target variable, which was `full_recovery_delays` was excluded from this, and the variable `partial_recovery_days` was also kept out of this to hopefully reduce the risk of any type of leakage when the predictors are interpreted at the end of the analysis.

The PCA results showed that the first principal component explained 12.13% of the variance, while all the first three principals together explained 23.93%. Another thing to take note of is that the dataset needed 16 components to explain about 80% of the variance and 19 components to explain about 90%. This means that the dataset contains many independent dimensions, and using only a few components doesn't encapsulate most of the variation in the data.

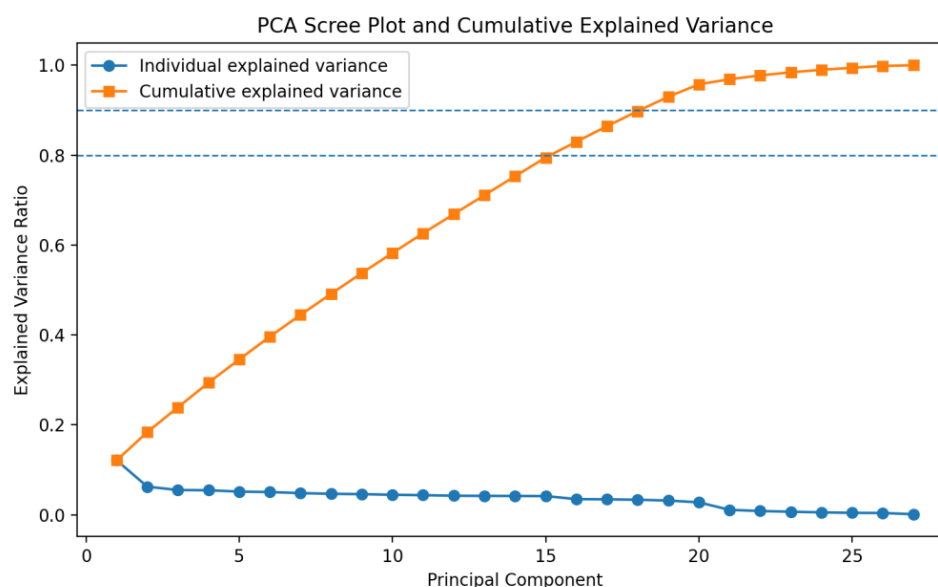


Figure 8: Scree plot for PCA and the cumulative explained variance

Table 2: PCA explained variance for the first five components

Component	Individual Variance That Is Explained	Cumulative Variance That Is Explained
PC1	12.13%	12.13%
PC2	6.27%	18.40%
PC3	5.53%	23.93%
PC4	5.47%	29.40%
PC5	5.17%	34.58%

The patterns from loading show that PC1 is for the most part correlated with the severity of the disruption, percentage of the production impact, severity_impact, and loss of revenue. This component can be seen as a disruption intensity dimension in a way. PC2 can be seen as the supplier size and availability of backup suppliers, which shows a resilience dimension for suppliers. PC3 can be seen as the supplier region variables, which shows a structure in the dataset for region or geographical location.

Discussions

Interpretation of Results

The results of this study helped accomplish the main goal of understanding what factors have the biggest impact on supply chain recovery times and how machine learning can be used to predict those outcomes. Through using exploratory data analysis, correlation analysis, and PCA, there were several important relationships that were found between disruption variables and recovery duration. The analysis showed that factors such as disruption severity, production impact percentage, response time, and loss of revenue all control how long it takes companies to fully recover from supply chain disruptions.

One of the largest findings from the analysis was that more severe disruptions usually have longer recovery periods. The heatmaps and boxplots showed a clear relationship between disruption severity and full recovery days, meaning that as the disruptions become more damaging, the companies need more time and resources to return to normal operations completely. This finding connects directly to the main project objective of identifying the variables that most strongly affect recovery duration. In a real business setting, this means companies should focus more attention on identifying highest risk disruptions early and creating plans that reduce damage before situations become worse.

The analysis also showed that different types of disruptions affect businesses in different ways. Cyberattacks and geopolitical disruptions tended to have longer, and more unpredictable recovery times compared to disruptions like port congestion. This suggests that some disruption events are much harder to manage because they are less predictable and often involve more uncertainty. The results imply that businesses should not rely on one universal recovery strategy. Instead, companies should build flexible response plans that are tailored to specific types of disruptions and industries.

Another important finding involved factors related to suppliers. The results showed that larger suppliers were more likely to have backup suppliers, and disruptions involving Tier 1 suppliers usually lead to greater financial losses and operational problems ultimately. This highlights how dependent many companies are on their primary suppliers. The findings suggest that businesses should invest in stronger supplier networks and develop backup supplier relationships to improve the resilience and strength of the supply chain. Relying too much on one supplier can increase risk and make recovery much more difficult during major disruptions.

The PCA results also helped show how complex supply chain disruptions really are. Since many of the principal components were needed to explain most of the variance in the dataset, the results show that recovery outcomes are affected by many different variables working together compared to only one factor by itself. The first principal component was connected to the actual disruption severity, as well as the production impact percentage, loss of revenue, and the impact the severity had, which altogether represented the overall intensity of a disruption. This supports the importance of using machine learning models that can handle these types of complicated relationships between variables.

Overall, the results of this project show that machine learning can be very useful for predicting supply chain recovery times as well as pinpointing the factors that contribute the most to how severe supply chain disruptions are. These findings can allow businesses to improve their strategies for risk management, help with better supplier planning, and help with responding to disruptions more effectively. By using predictive analytics like this, companies can make better decisions overall and reduce possible downtime, lower financial losses, and improve the strength of their supply chain.

Appendix

Table 1: Data Variables and Nature

#	Variable Name	Description	Feature Nature
*	full_recovery_days	Number of days to achieve full operational recovery (The target variable).	Target
0	disruption_id	Unique identifier for the disruption record.	Non-Predictive
1	partial_recovery_days	Number of days to achieve partial operational recovery (Excluded to prevent leakage).	Non-Predictive
2	disruption_type	Type of disruption event (e.g., Cyber Attack, Factory Incident, Labor Strike).	Event-based
3	industry	The industry sector of the affected supply chain (e.g., Electronics, Pharmaceuticals).	Operational
4	supplier_tier	Tier level of the supplier affected (1, 2, or 3).	Logistical
5	supplier_region	Geographic region where the supplier is located.	Logistical
6	supplier_size	Size category for supplier (Small, Medium, Large).	Logistical
7	has_backup_supplier	Indicator if a backup supplier is available (True/False or 0/1).	Logistical

8	disruption_severity	Severity score rating of the disruption event (Scale 1–5).	Event-based
9	production_impact_pct	Percentage reduction in production capability due to disruption.	Operational
10	revenue_loss_usd	Estimated financial revenue loss in USD resulting from the disruption.	Financial
11	response_type	Mitigation strategy implemented (e.g., Alternative Supplier, Customer Delay).	Operational
12	response_time_days	Time taken in days to execute the mitigation response strategy.	Operational
13	permanent_supplier_change	Indicator if the disruption led to a permanent change in vendor (True/False or 0/1).	Logistical
14	severity_impact	Interaction features combining disruption severity and production impact percentage.	Engineered

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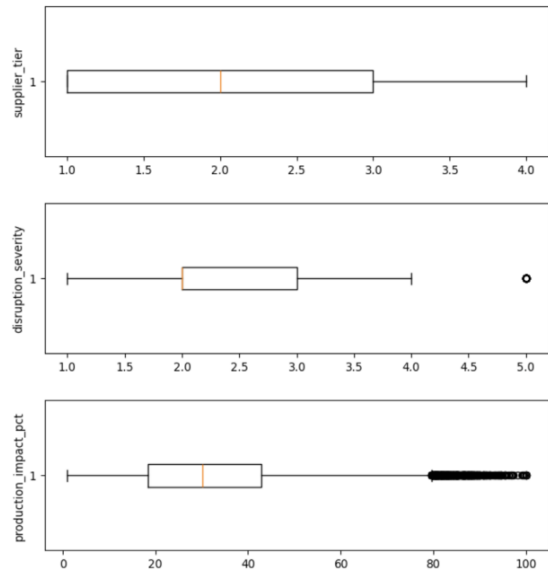


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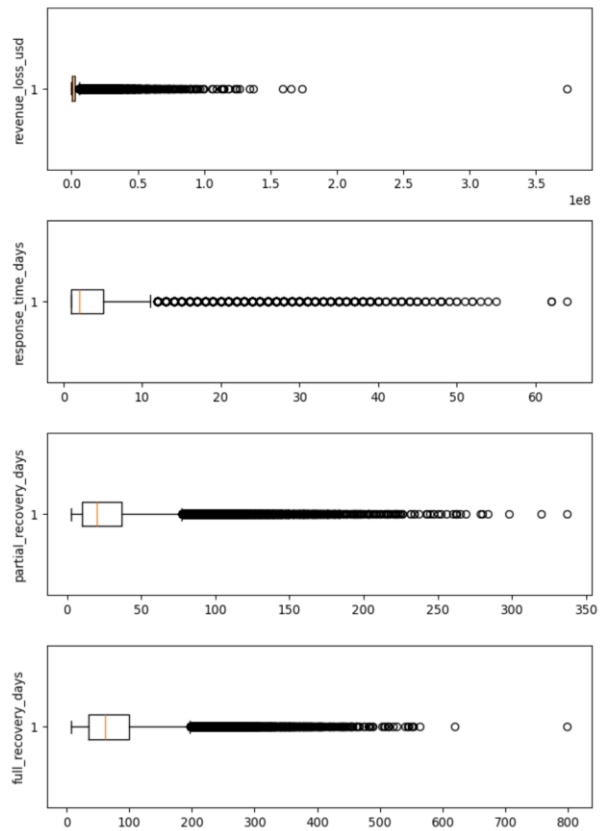


Figure 2: Revenue Loss, Response Time, Partial/Full Recovery Days Outliers Boxplots

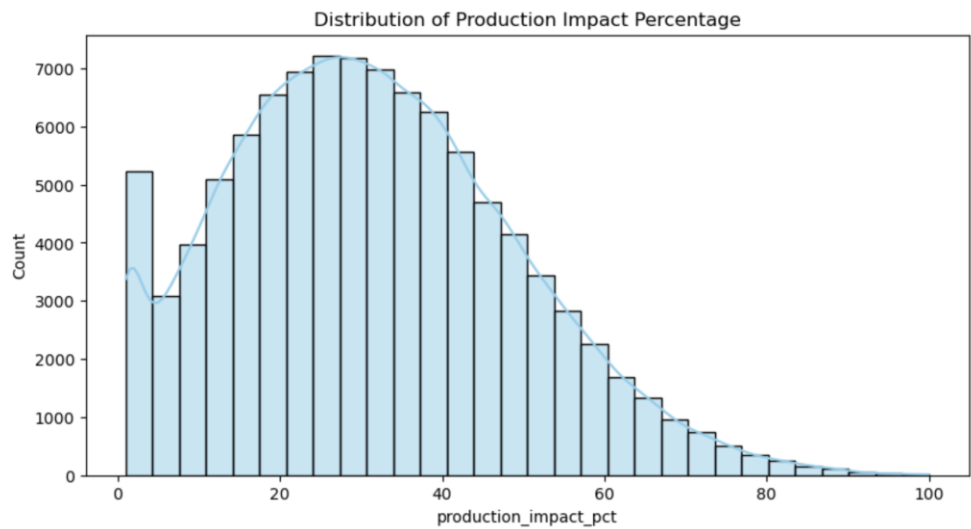


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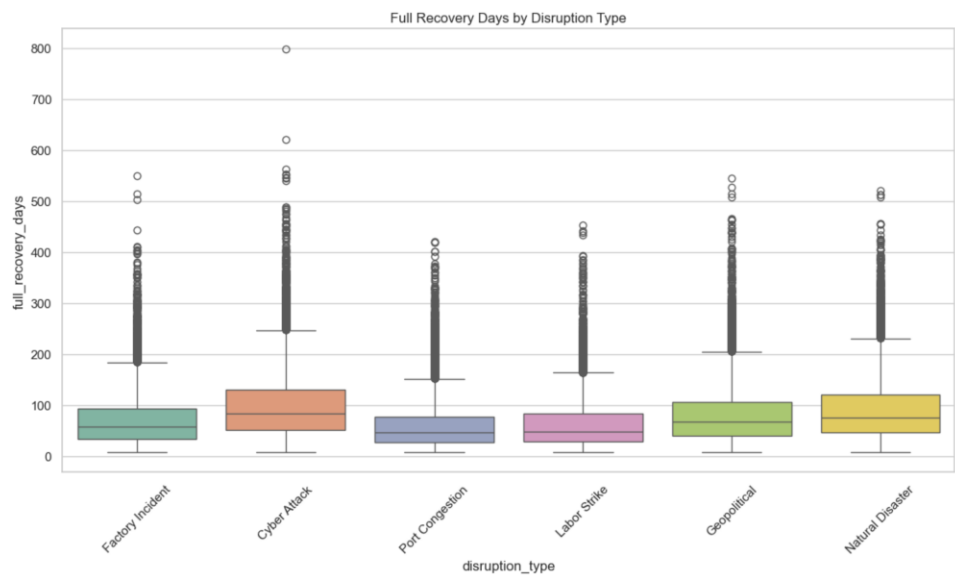


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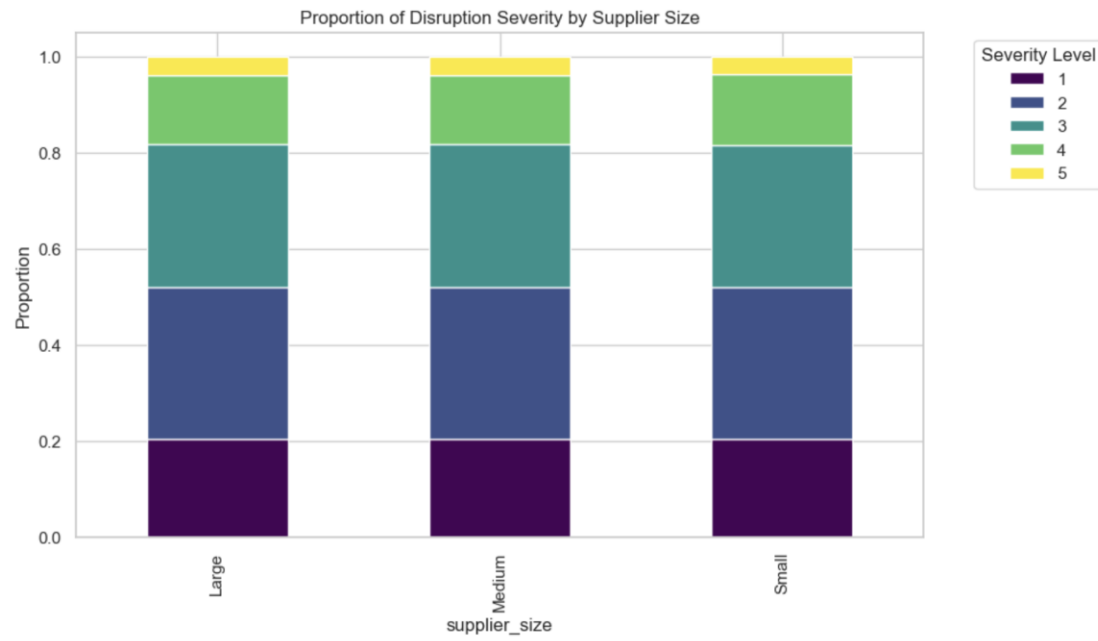


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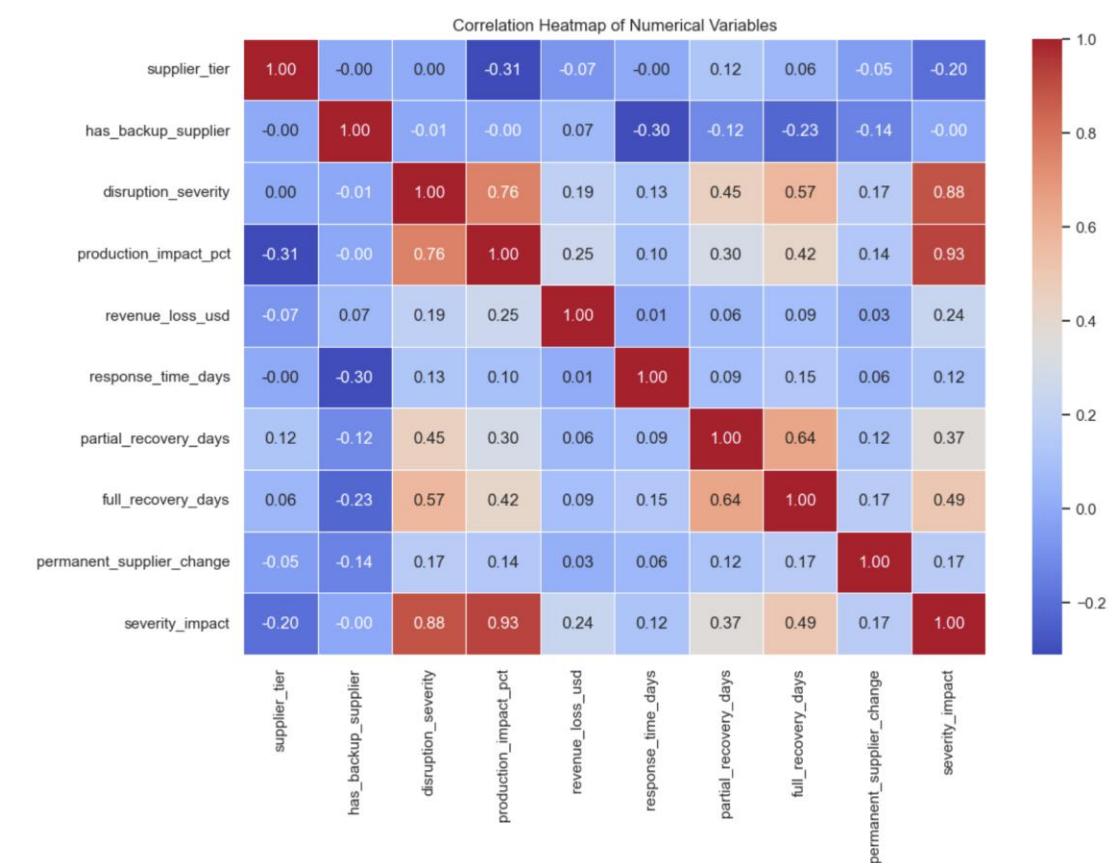


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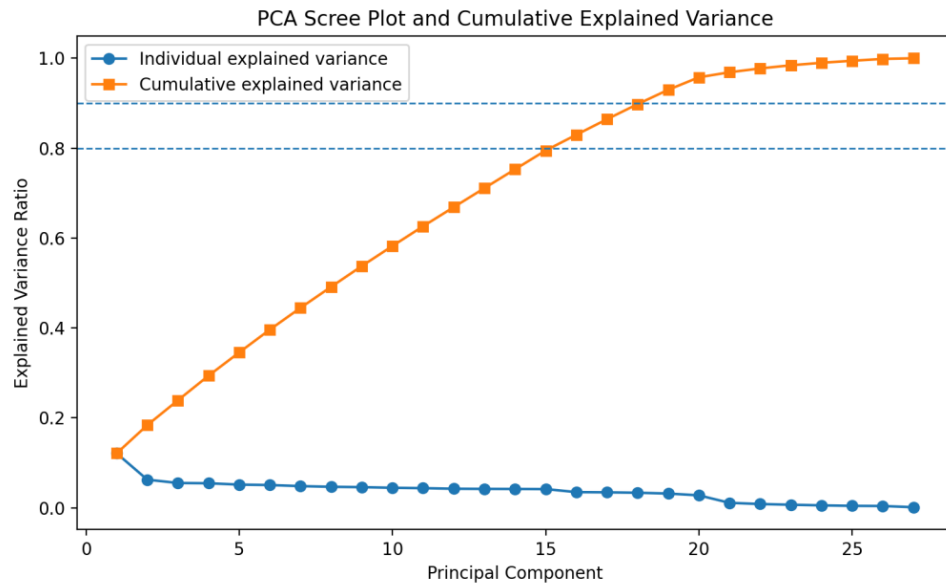


Figure 8: Scree plot for PCA and the cumulative explained variance

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